

springs, refer to the flow chart above to confirm formulas

- Calculate the alpha condition:

$$\alpha < 2.63 \frac{D}{L_0}$$

- Use this table to see how you can support the spring:

Table 10–2 End-Condition Constants α for Helical Compression Springs*

End Condition	Constant α
Spring supported between flat parallel surfaces (fixed ends)	0.5
One end supported by flat surface perpendicular to spring axis (fixed); other end pivoted (hinged)	0.707
Both ends pivoted (hinged)	1
One end clamped; other end free	2

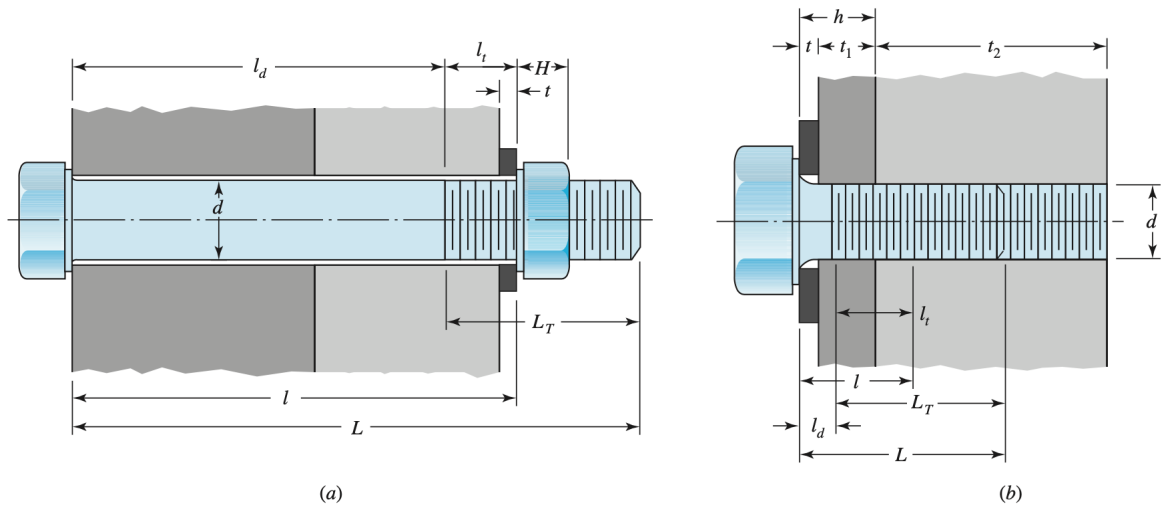
*Ends supported by flat surfaces must be squared and ground.

5.4 Fasteners/Bolts

5.4.1 Design Selection

1. Determine a suitable length for the bolt:

- (a) Compute the grip length.



- If your bolt arrangement is like figure (a):

l = thickness of all material squeezed between face of bolt and face of nut

- If your bolt arrangement is like figure (b):

$$l = \begin{cases} h + t_2/2, & t_2 < d \\ h + d/2 & t_2 \geq d \end{cases}$$

(b) Compute the fastener/bolt length (round up to nearest 1/4th inch):

- For figure (a):

$$L > l + H$$

- for figure (b):

$$L > h + 1.5d$$

2. Compute the bolt stiffness k_b :

(a) Determine the threaded length L_T :

- If your units are in inches:

$$L_T = \begin{cases} 2d + \frac{1}{4} \text{ in} & L \leq 6 \text{ in} \\ 2d + \frac{1}{2} \text{ in} & L > 6 \text{ in} \end{cases}$$

- If your units are in mm:

$$L_T = \begin{cases} 2d + 6 \text{ mm} & L \leq 125 \text{ mm}, d \leq 48 \text{ mm} \\ 2d + 12 \text{ mm} & 125 < L \leq 200 \text{ mm} \\ 2d + 25 \text{ mm} & L > 200 \text{ mm} \end{cases}$$

(b) Compute the length of the unthreaded portion in grip l_d :

$$l_d = L - L_T$$

(c) Compute the length of the threaded portion in grip l_t :

$$l_t = l - l_d$$

(d) Compute the area of the unthreaded portion A_d :

$$A_d = \pi d^2/4$$

(e) Find the area of the threaded portion A_t from one of the following tables (it is the tensile-stress area A_t):

Table 8–1 Diameters and Areas of Coarse-Pitch and Fine-Pitch Metric Threads*

Nominal Major Diameter d mm	Coarse-Pitch Series			Fine-Pitch Series		
	Pitch p mm	Tensile- Stress Area A_t mm^2	Minor- Diameter Area A_r mm^2	Pitch p mm	Tensile- Stress Area A_t mm^2	Minor- Diameter Area A_r mm^2
1.6	0.35	1.27	1.07			
2	0.40	2.07	1.79			
2.5	0.45	3.39	2.98			
3	0.5	5.03	4.47			
3.5	0.6	6.78	6.00			
4	0.7	8.78	7.75			
5	0.8	14.2	12.7			
6	1	20.1	17.9			
8	1.25	36.6	32.8	1	39.2	36.0
10	1.5	58.0	52.3	1.25	61.2	56.3
12	1.75	84.3	76.3	1.25	92.1	86.0
14	2	115	104	1.5	125	116
16	2	157	144	1.5	167	157
20	2.5	245	225	1.5	272	259
24	3	353	324	2	384	365
30	3.5	561	519	2	621	596
36	4	817	759	2	915	884
42	4.5	1120	1050	2	1260	1230
48	5	1470	1380	2	1670	1630
56	5.5	2030	1910	2	2300	2250
64	6	2680	2520	2	3030	2980
72	6	3460	3280	2	3860	3800
80	6	4340	4140	1.5	4850	4800
90	6	5590	5360	2	6100	6020
100	6	6990	6740	2	7560	7470
110				2	9180	9080

Table 8–2 Diameters and Area of Unified Screw Threads UNC and UNF*

Size Designation	Nominal Major Diameter in	Coarse Series—UNC			Fine Series—UNF		
		Threads per Inch N	Tensile-Stress Area A_t in ²	Minor-Diameter Area A_r in ²	Threads per Inch N	Tensile-Stress Area A_t in ²	Minor-Diameter Area A_r in ²
0	0.0600				80	0.001 80	0.001 51
1	0.0730	64	0.002 63	0.002 18	72	0.002 78	0.002 37
2	0.0860	56	0.003 70	0.003 10	64	0.003 94	0.003 39
3	0.0990	48	0.004 87	0.004 06	56	0.005 23	0.004 51
4	0.1120	40	0.006 04	0.004 96	48	0.006 61	0.005 66
5	0.1250	40	0.007 96	0.006 72	44	0.008 80	0.007 16
6	0.1380	32	0.009 09	0.007 45	40	0.010 15	0.008 74
8	0.1640	32	0.014 0	0.011 96	36	0.014 74	0.012 85
10	0.1900	24	0.017 5	0.014 50	32	0.020 0	0.017 5
12	0.2160	24	0.024 2	0.020 6	28	0.025 8	0.022 6
$\frac{1}{4}$	0.2500	20	0.031 8	0.026 9	28	0.036 4	0.032 6
$\frac{5}{16}$	0.3125	18	0.052 4	0.045 4	24	0.058 0	0.052 4
$\frac{3}{8}$	0.3750	16	0.077 5	0.067 8	24	0.087 8	0.080 9
$\frac{7}{16}$	0.4375	14	0.106 3	0.093 3	20	0.118 7	0.109 0
$\frac{1}{2}$	0.5000	13	0.141 9	0.125 7	20	0.159 9	0.148 6
$\frac{9}{16}$	0.5625	12	0.182	0.162	18	0.203	0.189
$\frac{5}{8}$	0.6250	11	0.226	0.202	18	0.256	0.240
$\frac{3}{4}$	0.7500	10	0.334	0.302	16	0.373	0.351
$\frac{7}{8}$	0.8750	9	0.462	0.419	14	0.509	0.480
1	1.0000	8	0.606	0.551	12	0.663	0.625
$1\frac{1}{4}$	1.2500	7	0.969	0.890	12	1.073	1.024
$1\frac{1}{2}$	1.5000	6	1.405	1.294	12	1.581	1.521

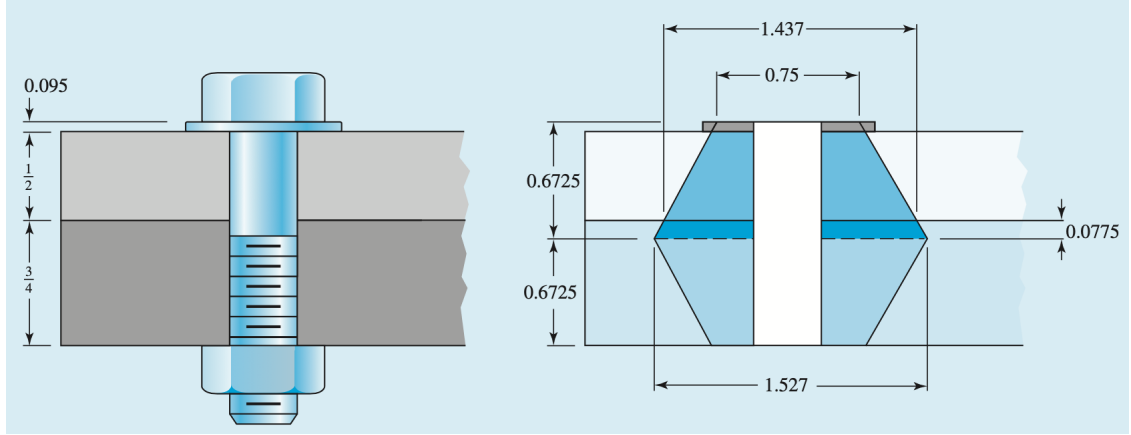
(f) Compute the bolt stiffness k_b (in lbf/in or N/m):

$$k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d}$$

where E is Young's Modulus

3. Determine the stiffness of the members:

- (a) Make a diagram like the one below. The angle at which the line goes down is assumed to be 30° . The line will start at a diameter of 1.5 times the nominal diameter of the bolt apart. The lines will start on the top washer (or if there is none then the top of the upper plate) and will end either on the bottom of the lower plate or if there is a second washer then into it as well. The lines will continue to separate until they reach the middle of the height of the region that they are allowed in. After that, they will start to converge again.



- (b) For each section of the diagram, (i.e. top washer, top plate, bottom plate, bottom washer etc.) calculate the spring rate k using the following expression:

$$k = \frac{0.5774\pi E d}{\ln \frac{(1.155t + D - d)(D + d)}{(1.155t + D + d)(D - d)}}$$

where E is Young's Modulus of the material,

d is the inner diameter of the section member

for plates, it is usually the nominal diameter of the bolt

for washers, it is their inner diameter

D is the shortest distance between two separating lines in the member's region,

t is the thickness of the section

Note that if two members have the same material and the inner diameter can they can be treated as one member where the thickness is the sum of their individual thicknesses.

Also note that if the lines go from separating to converging in the middle of a member's region, then you have to treat the section where they are separating as a different member from the section where they are converging. This is because they will have a different D value.

- (c) Once you have calculated all the spring rates for the individual members you can add them with the following equation:

$$\frac{1}{k_t} = \frac{1}{k_1} + \dots + \frac{1}{k_n}$$

- (d) If the top half and bottom half are the same, then you can calculate the spring rate of the top half with:

$$\frac{1}{k_{top}} = \frac{1}{k_1} + \dots + \frac{1}{k_n}$$

and the total spring rate of the joint will be:

$$k_t = \frac{k_{top}}{2}$$

5.4.2 Factor of Safety and Preload Calculations

1. Determine the torque to reach a given preload

- (a) Calculate P , the load per bolt.

$$P = \frac{P_{tot}}{N}$$

Where P_{tot} is the total load in the tension joint,

N is the number of bolts

- (b) Find the proof strength S_p , the minimum tensile strength S_{ut} and the endurance strength of the bolt S_e from the following tables:

Table 8–9 SAE Specifications for Steel Bolts

SAE Grade No.	Size Range Inclusive, in	Minimum Proof Strength,* kpsi	Minimum Tensile Strength,* kpsi	Minimum Yield Strength,* kpsi	Material	Head Marking
1	$\frac{1}{4}$ – $1\frac{1}{2}$	33	60	36	Low or medium carbon	
2	$\frac{1}{4}$ – $\frac{3}{4}$ $\frac{7}{8}$ – $1\frac{1}{2}$	55 33	74 60	57 36	Low or medium carbon	
4	$\frac{1}{4}$ – $1\frac{1}{2}$	65	115	100	Medium carbon, cold-drawn	
5	$\frac{1}{4}$ –1 $1\frac{1}{8}$ – $1\frac{1}{2}$	85 74	120 105	92 81	Medium carbon, Q&T	
5.2	$\frac{1}{4}$ –1	85	120	92	Low-carbon martensite, Q&T	
7	$\frac{1}{4}$ – $1\frac{1}{2}$	105	133	115	Medium-carbon alloy, Q&T	
8	$\frac{1}{4}$ – $1\frac{1}{2}$	120	150	130	Medium-carbon alloy, Q&T	
8.2	$\frac{1}{4}$ –1	120	150	130	Low-carbon martensite, Q&T	

Table 8–10 ASTM Specifications for Steel Bolts

ASTM Designation No.	Size Range, Inclusive, in	Minimum Proof Strength,* kpsi	Minimum Tensile Strength,* kpsi	Minimum Yield Strength,* kpsi	Material	Head Marking
A307	$\frac{1}{4}$ – $1\frac{1}{2}$	33	60	36	Low carbon	
A325, type 1	$\frac{1}{2}$ –1 $1\frac{1}{8}$ – $1\frac{1}{2}$	85	120	92	Medium carbon, Q&T	
		74	105	81		
A325, type 2	$\frac{1}{2}$ –1 $1\frac{1}{8}$ – $1\frac{1}{2}$	85	120	92	Low-carbon, martensite, Q&T	
		74	105	81		
A325, type 3	$\frac{1}{2}$ –1 $1\frac{1}{8}$ – $1\frac{1}{2}$	85	120	92	Weathering steel, Q&T	
		74	105	81		
A354, grade BC	$\frac{1}{4}$ – $2\frac{1}{2}$ $2\frac{3}{4}$ –4	105	125	109	Alloy steel, Q&T	
		95	115	99		
A354, grade BD	$\frac{1}{4}$ –4	120	150	130	Alloy steel, Q&T	
A449	$\frac{1}{2}$ –1 $1\frac{1}{8}$ – $1\frac{1}{2}$ $1\frac{3}{4}$ –3	85	120	92	Medium-carbon, Q&T	
		74	105	81		
		55	90	58		
A490, type 1	$\frac{1}{2}$ – $1\frac{1}{2}$	120	150	130	Alloy steel, Q&T	
A490, type 3	$\frac{1}{2}$ – $1\frac{1}{2}$	120	150	130	Weathering steel, Q&T	

Table 8–11 Metric Mechanical-Property Classes for Steel Bolts, Screws, and Studs

Property Class	Size Range, Inclusive	Minimum Proof Strength,* MPa	Minimum Tensile Strength,* MPa	Minimum Yield Strength,* MPa	Material	Head Marking
4.6	M5–M36	225	400	240	Low or medium carbon	
4.8	M1.6–M16	310	420	340	Low or medium carbon	
5.8	M5–M24	380	520	420	Low or medium carbon	
8.8	M16–M36	600	830	660	Medium carbon, Q&T	
9.8	M1.6–M16	650	900	720	Medium carbon, Q&T	
10.9	M5–M36	830	1040	940	Low-carbon martensite, Q&T	
12.9	M1.6–M36	970	1220	1100	Alloy, Q&T	

Table 8–17 Fully Corrected Endurance Strengths for Bolts and Screws with Rolled Threads*

Grade or Class	Size Range	Endurance Strength
SAE 5	$\frac{1}{4}$ –1 in	18.6 kpsi
	$1\frac{1}{8}$ – $1\frac{1}{2}$ in	16.3 kpsi
SAE 7	$\frac{1}{4}$ – $1\frac{1}{2}$ in	20.6 kpsi
SAE 8	$\frac{1}{4}$ – $1\frac{1}{2}$ in	23.2 kpsi
ISO 8.8	M16–M36	129 MPa
ISO 9.8	M1.6–M16	140 MPa
ISO 10.9	M5–M36	162 MPa
ISO 12.9	M1.6–M36	190 MPa

(c) Determine the initial bolt tension/preload force:

- If given there you are done.
- If said to be X% of proof strength, then the preload force F_i is

$$F_i = \frac{X}{100} A_t S_p$$

where A_t is the tensile-stress area given in Table 8-2

S_p is the proof strength of the bolt

- If the question asks you to choose then from Shigley's recommendation:

For nonpermanent connections, reused fasteners: $F_i = .75 A_t S_p$

For permanent connections: $F_i = .90 A_t S_p$

(d) Choose a torque factor K from the table

Bolt Condition	K
Nonplated, black finish	0.30
Zinc-plated	0.20
Lubricated	0.18
Cadmium-plated	0.16
With Bowman Anti-Seize	0.12
With Bowman-Grip nuts	0.09

(e) Compute the torque from the given formula:

$$T = K F_i d$$

where T is the torque, K is the torque factor,

F_i is the initial bolt tension,

d is the nominal diameter of the bolt

- Note that according to the homework solutions a torque of around 1000 lb-in is "VERY high" for a wrench

2. Compute overload/load factor of safety

- (a) Calculate the stiffness constant of the joint:

$$C = \frac{k_b}{k_b + k_m}$$

Where k_b is the bolt spring rate,

k_m is the member spring rate

Note that the stiffness constant of the joint represents what percent of the load is picked up by the bolt compared to the members.

- (b) Calculate the load factor:

$$n_L = \frac{S_P A_t - F_i}{CP}$$

Where S_P is the proof strength of the bolt in Pa or psi,

A_t is the tensile-stress area given in Table 8-1 or 8-2 above in m^2 or in^2 ,

F_i is the initial bolt tension in N or lb,

C is the stiffness constant,

P is the tension load per bolt in N or lb

Note that to convert from mm^2 to m^2 , multiply by 10^6

3. Compute bolt yield/yielding factor of safety:

- (a) The tensile stress in the bolt is:

$$\sigma_b = \frac{F_b}{A_t} = \frac{CP + F_i}{A_t}$$

Where A_t is the tensile-stress area given in Table 8-1 or 8-2 above in m^2 or in^2 ,

F_i is the initial bolt tension in N or lb,

C is the stiffness constant,

P is the tension load per bolt in N or lb

- (b) The static yielding factor of safety is:

$$n_p = \frac{S_p}{\sigma_b} = \frac{S_p A_t}{CP + F_i}$$

4. Compute the joint separation factor of safety:

$$n_0 = \frac{F_i}{P(1 - C)}$$

5. Compute the Goodman criteria fatigue factor of safety:

- (a) Compute the alternating stress:

$$\sigma_a = \frac{C(P_{max} - P_{min})}{2A_t}$$

(b) Compute the midrange stress:

$$\sigma_m = \frac{C(P_{max} + P_{min})}{2A_t} + \frac{F_i}{A_t}$$

(c) Compute the initial stress:

$$\sigma_i = \frac{F_i}{A_t}$$

(d) Compute the fatigue factor of safety:

$$n_f = \frac{S_e(S_{ut} - \sigma_i)}{S_{ut}\sigma_a + S_e(\sigma_m - \sigma_i)}$$

Where S_e is the endurance strength,

S_{ut} is the tensile strength,

σ_i is the initial stress,

σ_m is the midrange stress,

σ_a is the alternating stress

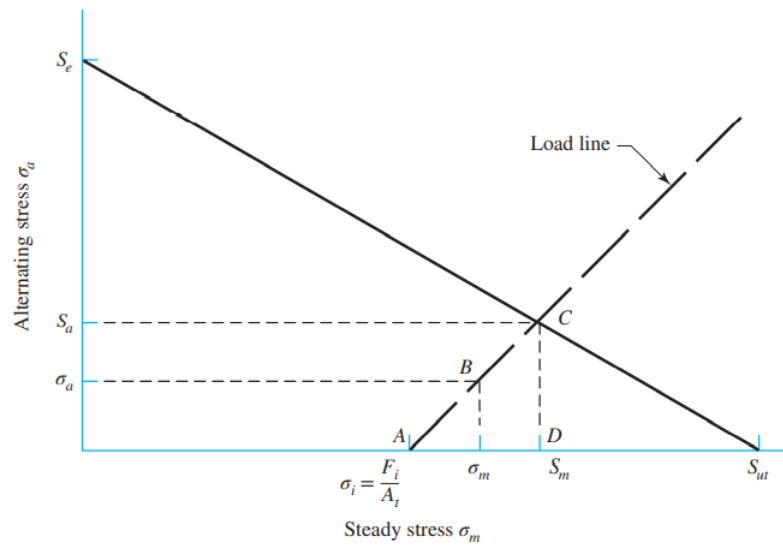
(e) If the external load is a repeated load i.e. (from 0 to P) then:

$$\sigma_a = \frac{CP}{2A_t}$$

$$\sigma_m = \frac{CP}{2A_t} + \frac{F_i}{A_t} = \sigma_a + \sigma_i$$

$$n_f = \frac{S_e(S_{ut} - \sigma_i)}{\sigma_a(S_{ut} + S_e)}$$

(f) If fatigue failure diagram is needed, it is shown below:



- The load line is given by the equation:

$$S_a = \frac{\sigma_a}{\sigma_m - \sigma_i}(S_m - \sigma_i)$$

- Then the Goodman line a.k.a. the line going from S_e to S_{ut} is given by the equation:

$$S_a = S_e - \frac{S_e}{S_{ut}}S_m$$

- Note that even though the graph has axes σ_a v.s. σ_m , it is essentially a S_a v.s. S_m graph. So just plot S_a as a function of S_m .
6. If any of the factor safeties are low or below 1, then the grade, size, or quantity of bolts can be increased to increase the safety factors.